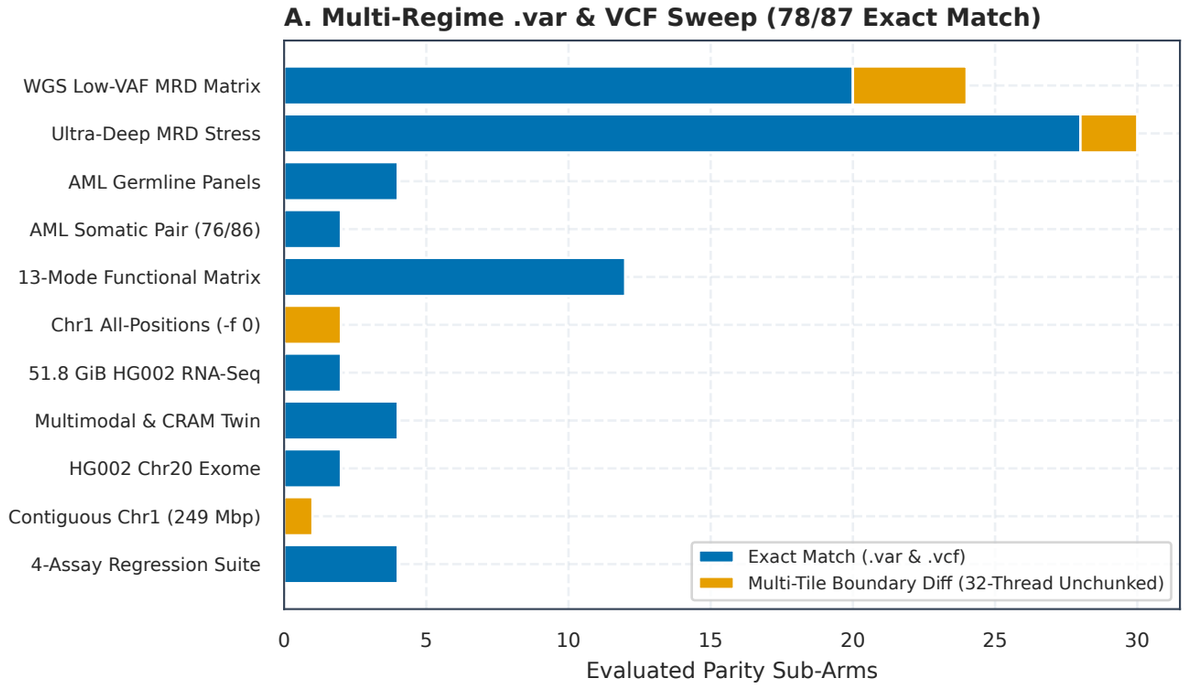
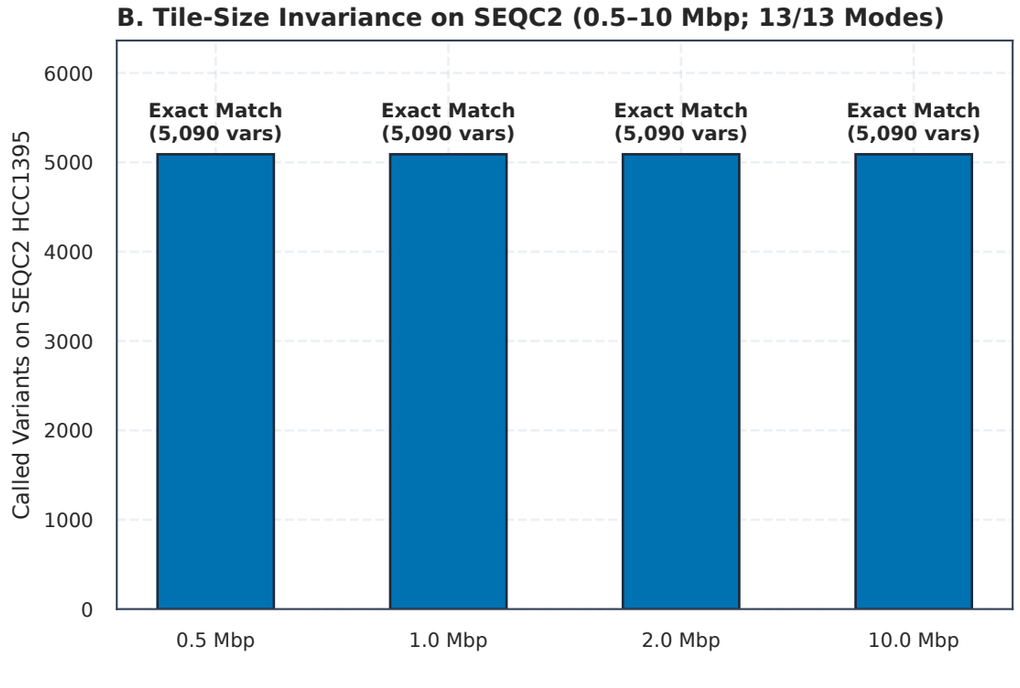
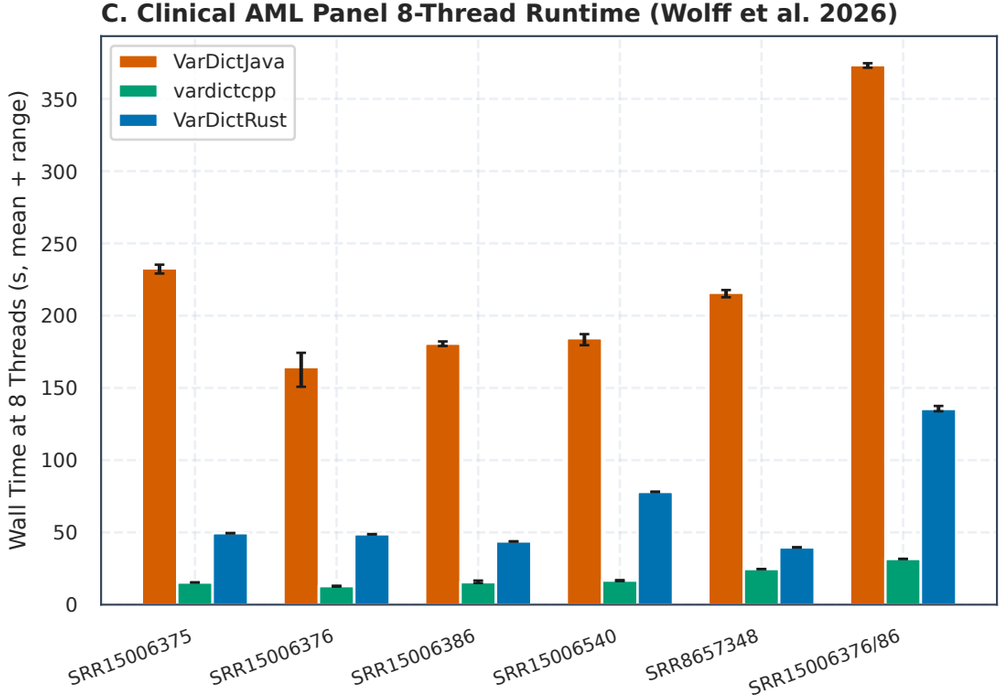
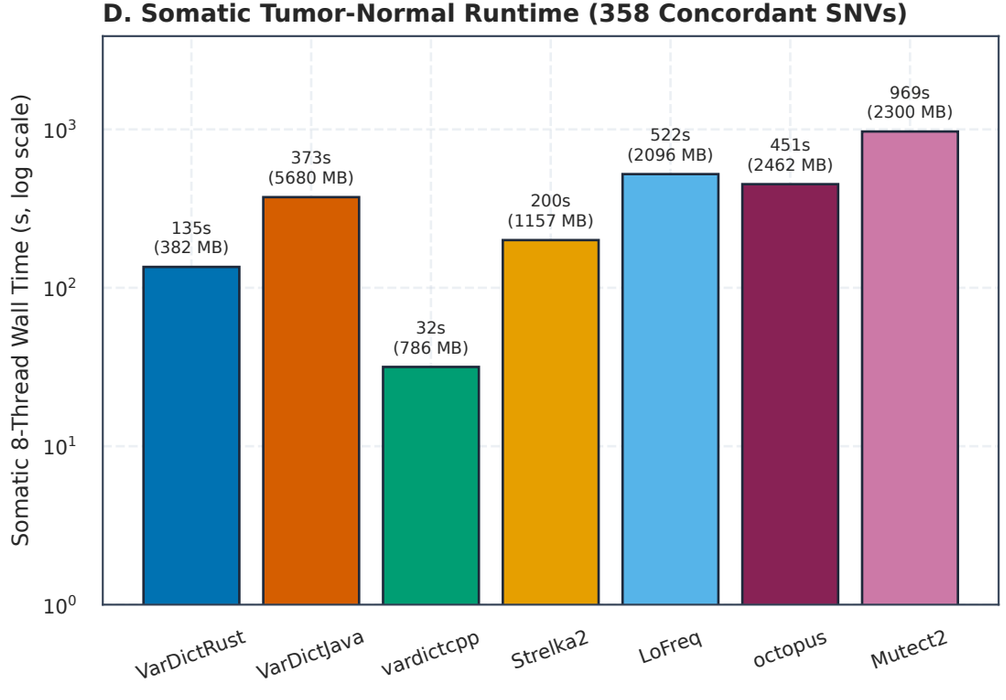

A. Multi-Regime .var & VCF Sweep (78/87 Exact Match)
WGS Low-VAF MRD Matrix
Ultra-Deep MRD Stress
AML Germline Panels
AML Somatic Pair (76/86)
13-Mode Functional Matrix
Chr1 All-Positions (-f 0)
51.8 GiB HG002 RNA-Seq
Multimodal & CRAM Twin
HG002 Chr20 Exome
Contiguous Chr1 (249 Mbp)
4-Assay Regression Suite
Exact Match (.var & .vcf)
Multi-Tile Boundary Diff (32-Thread Unchunked)
0
5
10
15
20
25
30
Evaluated Parity Sub-Arms


B. Tile-Size Invariance on SEQC2 (0.5–10 Mbp; 13/13 Modes)
Called Variants on SEQC2 HCC1395
0
1000
2000
3000
4000
5000
6000
Exact Match (5,090 vars)
Exact Match (5,090 vars)
Exact Match (5,090 vars)
Exact Match (5,090 vars)
0.5 Mbp
1.0 Mbp
2.0 Mbp
10.0 Mbp


C. Clinical AML Panel 8-Thread Runtime (Wolff et al. 2026)
VarDictJava
vardictcpp
VarDictRust
Wall Time at 8 Threads (s, mean + range)
0
50
100
150
200
250
300
350
SRR15006375
SRR15006376
SRR15006386
SRR15006540
SRR8657348
SRR15006376/86


D. Somatic Tumor-Normal Runtime (358 Concordant SNVs)
Somatic 8-Thread Wall Time (s, log scale)
10^0
10^1
10^2
10^3
135s (382 MB)
373s (5680 MB)
32s (786 MB)
200s (1157 MB)
522s (2096 MB)
451s (2462 MB)
969s (2300 MB)
VarDictRust
VarDictJava
vardictcpp
Strelka2
LoFreq
octopus
Mutect2